Zhaoyang Li

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Education

PhD Rutgers University–New Brunswick Sep 2026 - Present
- Computer Science; Robot Learning Lab

MS University of California San Diego Sep 2023 - Jun 2025
- Electrical and Computer Engineering (Intelligent Systems, Robotics & Control)

BS University of Wisconsin-Madison, Jan 2021 - May 2023
- **Double Major** in Computer Science & Mathematics

Research Interests

My research focuses on developing embodied agents that perceive, reason, and act in the physical world. I work at the intersection of **multimodal learning**, **robot learning**, and **world models** with a focus on **(1).** improving the robustness of vision–language models and their deployment in robotics, **(2).** developing efficient control and planning policies, and **(3).** learning structured, object-centric world models that support prediction and planning. Ultimately, I aim to build reliable, interpretable agents that generalize to real-world environments.

Publications & Preprints

* indicates equal contributions.

- **Zhaoyang Li***, Aleyna Kara*, Abha Jha*, Bo-Ruei Huang, Majid Khadiv, Erdem Bıyık. "BFN Policy: Visuomotor Policy Learning for Hybrid Action Spaces via Bayesian Flow Networks.", Submitted to the Conference on Robot Learning **(CoRL)**, 2026.
- **Zhaoyang Li***, Zhan Ling*, Yuchen Zhou, Litian Gong, Erdem Bıyık, Hao Su. "ORIC: Benchmarking Object Recognition under Contextual Incongruity in Large Vision-Language Models.", in Proceedings of the IEEE / CVF Computer Vision and Pattern Recognition Conference **(CVPR)**, 2026. [Paper] [Code]
- Tongzhou Mu*, **Zhaoyang Li***, Stanisław Wiktor Strzelecki*, Xiu Yuan, Yunchao Yao, Litian Liang, Aditya Gulati, Hao Su. "When Should We Prefer State-to-Visual DAgger Over Visual Reinforcement Learning?", in Proceedings of the AAAI Conference on Artificial Intelligence **(AAAI)**, 2025. [Paper] [Code]

Research Experience & Project

Bayesian Flow Networks for Robotic Manipulation (Project Lead)
Advisor: Prof. Erdem Bıyık, USC — Dec 2025 – Apr 2026, Los Angeles, CA
- Developed BFN Policy, a Bayesian Flow Network–based imitation learning method that natively models hybrid (discrete–continuous) action spaces, avoiding the relaxation-and-rounding that breaks diffusion policies.
- Outperformed diffusion baselines on 7 of 8 parameterized-action benchmarks (robust as discrete dimension grew) at $\sim 8\times$ faster inference; validated on real-robot Push-T (xArm, WidowX). First-author paper submitted to CoRL 2026.

Context-Incongruity Robustness for Large Vision Language Models (Project Lead) Aug 2024 - Sep 2025
Advisor: Prof. Hao Su (UC San Diego); Prof. Erdem Bıyık (USC) — San Diego, CA
- Built ORIC to generate context-incongruous object–context data via dual LLM- and CLIP-guided sampling, and released the ORIC-Bench benchmark; **published at CVPR 2026**.
- Benchmarked 18 LVLMs and 2 open-vocabulary detectors, exposing systematic recognition bias under contextual incongruity.
- Fine-tuned Qwen3-VL-8B-Instruct with Visual-RFT on 600 ORIC samples, boosting F1 and human-aligned predictions on ORIC-Bench, AMBER, and HallusionBench.

Multi-modal Language Model for Drug Interaction Prediction (Project Lead) Dec 2024 - May 2025
Advisor: Prof. Pengtao Xie; UC-San Diego San Diego, CA

- Fine-tuned a multi-modal LLM with SMILES inputs to predict drug interaction status, degree, and mechanisms, integrating chemical informatics and NLP.
- Achieved strong performance: METEOR 0.42, BLEU-1 0.25, semantic similarity 0.57; outperforming GPT-4o (METEOR 0.16, BLEU-1 0.11, semantic similarity 0.30).

Empirical Analysis of State-to-Visual (S2V) Imitation vs. Visual RL (Project Co-Lead) Feb 2024 - Sep 2024
Advisor: Prof. Hao Su, UC-San Diego San Diego, CA

- Benchmarked State-to-Visual DAgger vs. visual RL across 16 tasks from ManiSkill, DMControl, and Adroit; **published at AAAI 2025**.
- Showed S2V DAgger excels on hard tasks with more consistent performance and lower wall-clock time, with no universal winner across tasks.
- Built a standardized S2V pipeline and derived practical recommendations for visual policy learning.

Simulation of the Connected and Automated Driving Systems Sep 2021 - May 2022
Advisor: Prof. Bin Ran; The Connected Automated Vehicle Highway System Group Madison, WI

- Simulated and optimized traffic systems in CARLA, enhancing traffic management models for improved efficiency.
- Refined object detection algorithms, including YOLO and Faster R-CNN, to improve vehicle detection and traffic control systems.

Teaching

Teaching Assistant at UW-Madison Spring 2023

- CS540: Introduction to Artificial Intelligence

Peer Mentor at UW-Madison Fall 2022

- CS537: Introduction to Operating System

Professional Services

Reviewer

- AAAI Conference on Artificial Intelligence
- AAAI Workshop: Large Language Models and Generative AI for Health

Technical Skills

- **Languages**: English (Proficient; TOEFL 108, GRE 330 + 3.5), Chinese (Native)
- **Programming**: Python, C++, C, Java, Matlab, R, LaTeX, SQL
- **ML and Vision Frameworks**: PyTorch, TensorFlow, OpenCV, scikit-learn, SimpleITK, SPM12, TorchIO
- **Simulation and Robotics**: CARLA, MuJoCo, ManiSkill, DMControl, RoboMimic, WidowX, Adroit.
- **Developer Tools**: VS Code, Vim, IntelliJ IDEA, Visual Studio, Git, Docker, Kubernetes